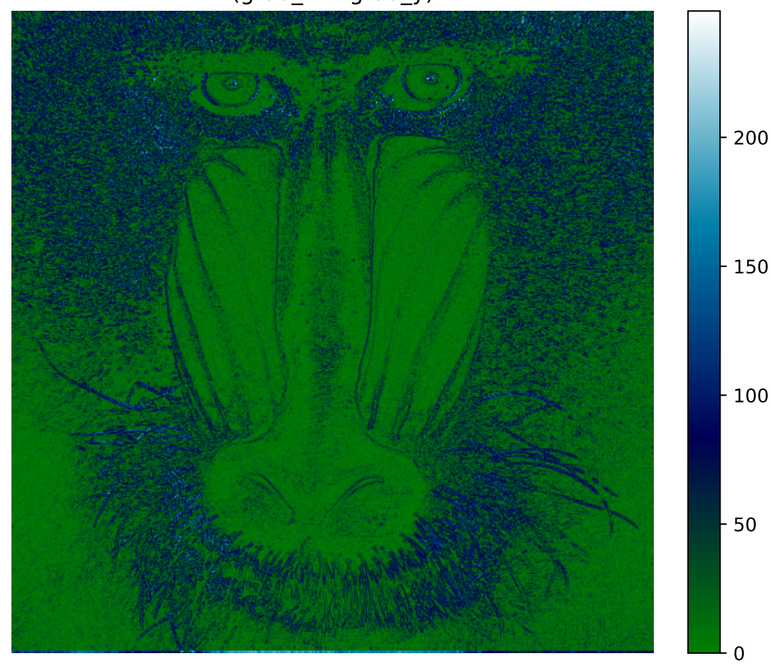
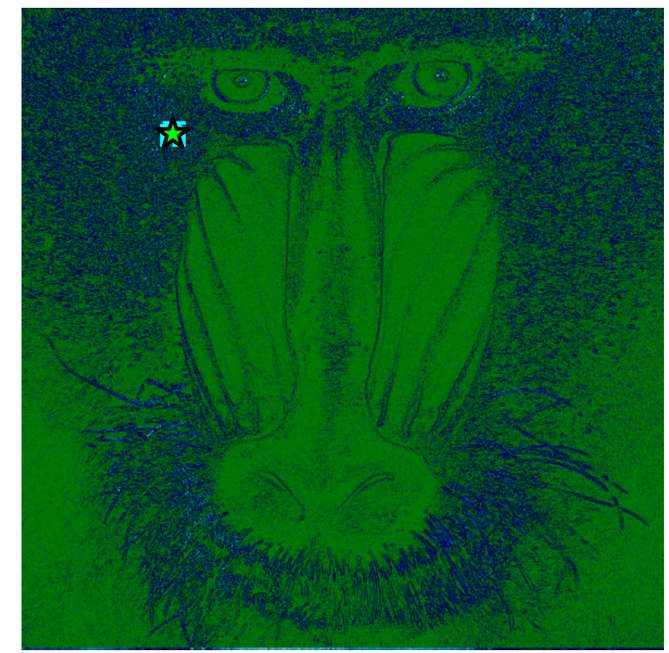
System Gradient Analysis - Actual Implementation
Phan Tich Gradient Theo He Thong Thuc Te

Original Image

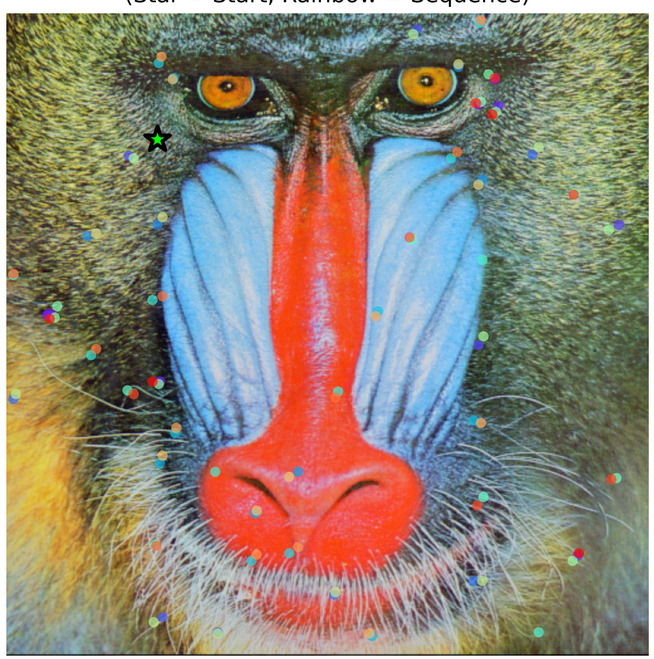
System Gradient Map
(grad_x + grad_y)



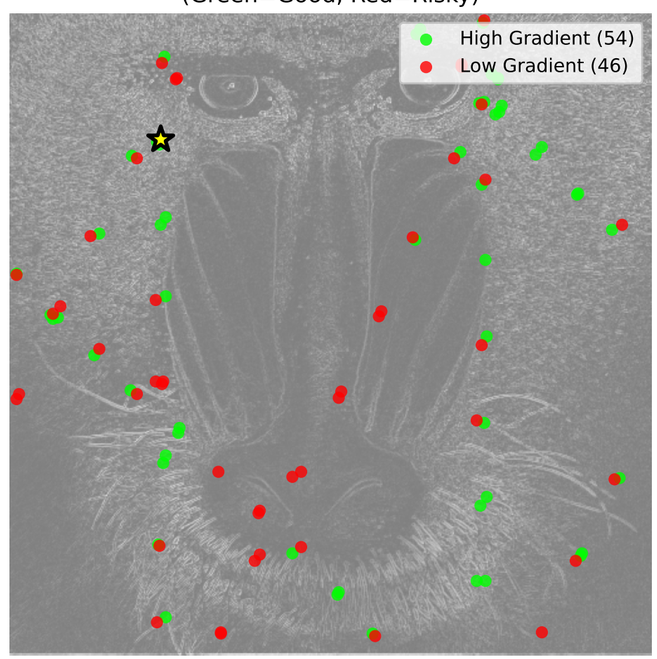
Starting Point (x0=120, y0=100)
Window: 16x16



First 100 Embedding Positions
(Star = Start, Rainbow = Sequence)



Position Quality Analysis
(Green=Good, Red=Risky)



Statistics & Quality Report

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SYSTEM ANALYSIS RESULTS
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IMAGE INFO:
- Size: 512 x 512 pixels
- Total pixels: 262,144

STARTING POINT (x0, y0):
- Position: (120, 100)
- Gradient value: 132.00
- Max gradient in image: 249.00
- Starting point quality: 53.0%

CHAOS PARAMETERS:
- Secret key: "default_key"
- Chaos key (int): 1928911999
- Algorithm: Arnold Cat Map + Logistic Map

EMBEDDING QUALITY (100 positions):
- High gradient (SAFE): 54 (54.0%)
- Low gradient (RISKY): 46 (46.0%)

CONCLUSION:
WARNING: Many positions in flat areas
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